



Universität zu Köln

Faculty of Arts and Humanities, Department of Linguistics

Master's Thesis

**Subject realisation in L2 Spanish:
An extended reproduction and replication of
Kocher (2025) with L1 German and L1 English
corpus data**

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree

Master of Arts

in the

Master Linguistik

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DATE OF SUBMISSION

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1 Introduction

- Spanisch als Nullsubjektsprache im Vergleich zu nicht-Nullsubjektsprachen Deutsch und Englisch
- Ziel der Arbeit ist Reproduktion von Kocher 2025 Paper + Erweiterung um die Gender und Genus Variablen + Ausweitung auf Englische L1 Lerner:innen des Spanischen
- Wieso dieses paper?
 - Highly relevant topic: Investigated which factors impact the differences in use between L1 and L2 Spanish
 - “Previous research into this topic was mainly based on experimentally elicited data, whereas Kocher applies an innovative statistical analysis procedure from learner corpus research” (Kocher, 2025: 28) (MuPDARF Modelling)
 - “Most of the studies furthermore distinguished exclusively between null and overt subjects, disregarding differences between DPs and pronouns, which, disregarding differences between DPs and pronouns. As a result, the interface contexts are very well understood, but more general properties and also the difference between DP and pronoun acquisition have been somewhat overlooked up until now” (Kocher, 2025: 28).
 - > Reproducing the original study ideally increases its confirmatory power.
 - Wieso Variable extension: No study has considered the role of the speaker’s gender or the genus of the referential subject in the context of second language (L2) Spanish null subject acquisition.
 - Comparing two different but closely related Germanic non-NSL might be interesting for investigate the nuances and insimilarities between them
- *We hope that our findings contribute to a deeper understanding of how Spanish learners with a non-null subject L1 acquire null subjects and overt pronouns.*
- *We also anticipate that our study will emphasise the need to enhance Open Science practices to improve the reproducibility and replicability of LCR.*
- Kapitelweise durch die Arbeit führen

2 Theoretical background

XXX

2.1 Subject realisation in Spanish

According to Chomsky languages can be classified as “pro-drop” or “non-pro-drop languages” (1993: 240). Meanwhile non-pro-drop languages, such as English and French, are characterised by the fact that they obligatorily use pronouns, in pro-drop languages, such as Italian and Spanish, overt pronominal subjects are not required (see Torres Cacoullos and Travis, 2019: 2). The Spanish language offers a rich inflectional system, in which verb forms are inflected for person, number, tense and aspect (see Ayoun, 2003: 82). Since the subject is indicated by verb itself, almost all Spanish verb forms are distinct and can therefore recover the omitted referential subject pronoun grammatically.

Thus, as depicted in example 1 Spanish has three main options to express referential subjects (see Kocher, 2025: 3):

- (1) (a) *Charles Chaplin está caminando por la calle.*
‘Charles Chaplin is walking down the street.’
(example ID 2_1, L1 Spanish)
- (b) *Él está caminando por la calle.*
‘He is walking down the street.’
(own example)
- (c) *Está caminando por la calle.*
‘[Charles Chaplin/ He] is walking down the street.’
(own example)

Either overt subjects can be used, such as *Charles Chaplin* in 1.a., or pronouns, such as *él*. Alternatively, covert or null subjects can be used, as in 1.c. In all three sentences, the subject refers to the same person. Embedded subject pronouns function the same way, as shown in example 2 (see Ayoun, 2003: 85)¹:

- (2) (a) *Al principio [Charles Chaplin] cree que el niño ha caído de un edificio.*
‘At first, [Charles Chaplin] thinks the boy has fallen from a building.’
(own example)
- (b) *Al principio [Charles Chaplin] cree que él ha caído de un edificio.*
‘At first, [Charles Chaplin] thinks he has fallen from a building.’
(own example)

¹For further pro-drop parameters, such as the impersonal expletive expressions or nominal subject inversion see Ayoun (2003): 85-86.

- (c) *Al principio [Charles Chaplin] cree que ha caído de un edificio.*
 ‘At first, [Charles Chaplin] thinks [the boy/he] has fallen from a building.’
 (example ID 12_4-5, L1 Spanish)

Zu pro-drop:

- Chomsky 1993 überfliegen
- Phinney lesen
-

Torres Cacoullos 2019:2 - In Spanish, rates of unexpressed 3sg subjects range from 97% in Pear Story narratives by Peninsular Spanish speakers (Comajoan 2006: 60, N = 285) to 73% in sociolinguistic interview narratives in Mexico City (Lastra and Butragueño 2015: 43, N = 450) and 68% in conversational Colombian Spanish (see Section 9, N = 1,431).

Omission rates differ between 54% and 80% depending on the variety of Spanish [Etxebarria Zuluaga (2022): XX].

Overt subjects, as in 1.a./2.a. and 1.b./2.b., are generally associated with [+referential shift] and [+focus], whereas null subjects, as in 1.c./2.c., express [+referential continuity] and thus the familiarity of the subject (see Etxebarria Zuluaga, 2022: 84). In order to further discuss these and other factors that influence the subject’s referential treatment in **?@sec-InfluentiaFactors**, the following section outlines the possible forms of subject realization in German and English.

- Referenzialität, Referentielle Kontinuität
- Informationsstruktur,
- Rolle von Satztyp
- Subjektposition

2.2 Comparing L1 systems: Subject realisation in German and English

- Subjektrealisierung im Deutschen, Subjektrealisierung im Englischen, Bedeutung für L2-Spanisch-Erwerb
- Referenzialität, Referentielle Kontinuität
- Informationsstruktur,
- Rolle von Satztyp
- Subjektposition
- VERGLEICHSTABELLE aller drei Sprachen!

Möglich in Englisch (Ayouun, 2003: 138)!!

German and English are non-NSLs that generally do not overt referential subjects (see [Ayoum, 2003](#): 84).

- Pro-Drop in German dialects und Umgangssprachlichen (haste, kannst, etc.) und mentale Verben (*glauben, denken, meinen, wissen*) (vgl. ids-Monnem dazu)

ENGLISH:

- Abgesehen von koreferenziellen Subjekten, deren Verben mit einer koordinierenden Konjunktion verbunden sind (z. B. *und, oder*), grenzt sich das Englische als nicht-Nullsubjektsprache insofern vom Spanischen ab, dass hier Nullsubjekte lediglich in prosodischen Anfangspositionen in deklarativen Satzstrukturen verwendet werden (see [Torres Cacoullos and Travis, 2019](#): 1) können. Dieser Restriktion unterliegt das Spanische nicht (vgl. [Torres Cacoullos and Travis, 2019](#) ebd.)
- z.B. Torres Cacoullos 2019:2 - In English spoken in the USA, rates of unexpressed vs. pronominal 3sg human specific subjects are five times higher in narratives than in conversation

2.3 Theoretical approaches to Foreign language learning

XXX

2.3.1 *Interface hypothesis*

- bspw. Sorace 2005, 2011; Tsimpli und Sorace 2006, etc

Nullsubjekte seien eher unproblematisch, während pronomina mit “permanent optionality and persistent difficulty” für Fremdsprachenlerner:innen des Spanischen assoziiert werden.

2.3.2 *Feature-Reassembly hypothesis*

- bspw. Lardiere 2008; Slabakova 2013

2.3.3 *Learnability* factors

- Transfer, Reassembly, Encoding in Kocher 2025
- See *Parameter Setting in Language Acquisition* Dalila Ayoun (IdSL I Sp2.2 895 p. 94/148f)

3 Prior research

XXX

3.1 Prior research on L1 German

- Zentrale Ergebnisse Kocher 2025, Vulnerabilität von Pronomen im Vgl. zu Null-Subjekten
- Weitere Studien und Methoden und Sprachdaten

3.2 Prior research on L1 English

- bspw. Liceras und Díaz 1999; Clements und Domínguez 2016; Domínguez und Arche 2022

3.3 Research Questions

RQ1: Sind die Ergebnisse aus Kocher (2025) im CEDEL2-Datensatz reproduzierbar?

RQ2: Haben das *Gender* der L2-Sprechenden und das Subjektgenus der Referent:innen im Korpus einen signifikanten Einfluss auf die Subjektrealisierung? Und tritt *Gender* dabei in signifikante Interaktionen mit anderen Prädiktoren?

RQ3: Inwiefern unterscheidet sich die Subjektrealisierung zwischen Sprecher:innen mit L1 Deutsch und L1 Englisch?

4 Methodology

- Fourfold study: (1) reproduction of original data, (2) robustness check of the annotation and L1 German speaker data results by extending the dataset, (3) close replication adding gender as a new variable to the model and another (4) close replication by applying the new model to L1 English speaker dataset.
- Name the subchapters

4.1 CEDEL2 Corpus

- Corpus Escrito del Español como L2
- The native Spanish speaker and the German and English speaker *absolvierten* the same task which does not allow very generalizable interpretation of the results. However, comparing the data of L1 speakers and L2 learners provides a precise and meaningful contrast.

4.2 Data

4.2.1 Data selection and pre-processing

First, assuring to be able to reproduce Kocher’s (2025) study results by rerunning their scripts and by using their data. Therefore, I used their manually annotated data for native Spanish and L1 German speakers extracting it from the study’s [OSF](#) repository, where I also accessed the R scripts my reproduction study is based on. The repository contains a `readme.md` file, which contains a summary of the files and dataframes, as well as brief information about the exclusion criteria, the key variables that entered into the final model, the names of the packages used, a short explanation about the analysis procedure used and finally, a recommendation on how to cite Kocher’s scripts and data. Also, it contains a folder with the processed and annotated native and L1 German speaker data, a R scripts with descriptive analysis (see `descriptive.R` in Appendix C), the MuPDAFRE analysis (see `mupdarfanalysis.R` in Appendix C) and another analysis in which the proficiency levels were calculated (see `demographics.R` in Appendix C). The created figures were saved in a separate folder within the repository.² The main reason for using Kocher’s L1 German speaker dataset instead of extracting the data from CEDEL2 again is that Kocher’s dataset has already been preprocessed and manually annotated, which facilitates the robustness check in the later study. Although Kocher’s L1 German dataset was based on the complete L1 German speech data from the *Charles Chaplin* task available on CEDEL2 at the time of her study, the native Spanish speaker dataset she used was based only on a selection of the available CEDEL2 data and contains 522 clauses.

After rerunning Kocher’s scripts with the original L1 German and Spanish native speaker data I aimed to test the robustness of Kocher’s (2025) results by adding more recent L1 German speaker data which was not yet included in the corpus when Kocher conducted their study. In total, I extracted 662 clauses from 17 additional L1 German speakers from CEDEL2 on 31 March 2026. The annotation filter in CEDEL2 ([Lozano, 2022](#)) criteria are displayed in Table 1:

²See [?@sec-Kocher](#) for an explanation of the scripts content.

Table 1: The corpus filter criteria for data extraction in CEDEL2. The remaining categories that are adjustable with a slider stayed in default mode (full range).

Corpus Filter Criteria	Selection
Result Type	Texts
Subcorpus	Learners of L2 Spanish
Medium	Written
Task Title	Chaplin
Sex, Proficiency, Filename	Any
L1	L1 German - L2 Spanish

Before the manual annotation the newly added L1 German data and the entire L1 English dataset were segmented into clause. Following Kocher (2025), subordinate clauses were separated from their corresponding main clauses and presented in the subsequent row. Within the main clause, omitted subordinate clauses were indicated by square brackets ([...]).

The annotation procedure largely followed the methodology established by Kocher (2025) in order to ensure comparability across datasets, still some additional information on the columns NSLHOME and NSLADDITIONAL are necessary, as the creation of these two variables was not explained in Kocher (2025). The variable NSLHOME was created based on the information provided in the *Languages.spoken.at.home* column and indicates whether a participant acquired a NSL in the home environment. In addition, the variable NSLADDITIONAL was derived from the FOREIGNLANGUAGE column and captures whether participants learned a NSL later through formal instruction or other means. For participants who had already acquired a NSL at home, NSLADDITIONAL was also coded as *Yes*, reflecting their continued exposure to a NSL. For participants without home exposure, NSLADDITIONAL was coded as *Yes* only if they had subsequently learned a NSL. For example, in the newly added L1 German dataset, only one participant had learned Italian as an additional language, whereas all remaining participants had learned only English, a non-NSL.³ Finally I joined the pre-processed L1 German dataset with the already annotated original L1 German data from Kocher (2025) and obtained a dataset of 2997 clauses.

In order to also check on the robustness of Kocher’s annotation I further extracted a data sample of `nrow(sample_kocher_germanL1)` clauses of three L1 German speakers by setting a seed, to ensure that the outcome of the randomly generated number series are always exactly the same, and re-annotating them (see Section 5.2.1 for the results).

For conducting the second close replication study comparing subject realisation in English L1 learner Spanish I used the same filter criteria as for the German L1 data extraction except changing the L1 learner group to “L1 English - L2 Spanish”. After extracting the dataset I segmented it so that each clause was displayed in one row. I used the machine-based tool *Meta*

³The more detailed pre-annotation process is explained in a separate .qmd file in Appendix C (see 1_Preprocessing.qmd).

Llama 3.3 70B Instruct and *DeepSeek R1 Distill LLama 70B* for this process via the University of Cologne’s academic cloud access. Then, I added the NLSADDITIONAL and NLSHOME columns and selected a data sample of 1300 clauses.

4.2.2 Exclusion criteria and data annotation

The annotation procedure largely followed the methodology established by Kocher (2025) in order to ensure comparability across datasets, while introducing a number of additional exclusion criteria. In general, Kocher’s included only sentences with referential subjects, i.e. declarative and interrogative personal sentences, whereas imperative verb forms (e.g., example (1)) and impersonal sentences, such as indefinite *se* constructions, e.g. example (2), or impersonal sentences (e.g., example (3)) were excluded and the corresponding clauses deleted (see Kocher (2025): 15). I decided to not only skip the annotation in these cases but to also delete the excluded clauses so that they won’t be accidentally selected in the random sample of 1300 clauses I base my L1 English data analysis on. Meanwhile, impersonal forms like example (4) were retained as they do not contain imperative forms.

1. [...] *que dice “por favor ama y cuida ese huérfano”* (datal2german, DE_WR_43_29_7_14_KP, excluded)
2. *En el video se ve un hombre* (datal2german, DE_WR_30_28_2_14_GV, excluded)
3. *Hay un infffantil* (datal2english, EN_WR_14_23_1_14_B, excluded)
4. *y le arrojaron basura* (datal2english, EN_WR_21_19_2_14_JB, ID35_2)
5. [...] *que dice: [Please love and take care of this orphan Child.]* (datal2german, DE_WR_27_19_1_14_AB, ID9_28)

Further, clauses containing English direct speech were excluded but not entirely deleted (see example (5)), whereas Spanish direct speech was retained (see also example (1) above).

Also, clauses without a verb (e.g. example (6)), clauses containing English lexical items within otherwise Spanish utterances (e.g. example (7)), and clauses in which the intended verb could not be identified (e.g. example (8)) were excluded from the annotation.

6. *¿Quizás la madre de otra bebé?* (datal2english, EN_WR_32_59_1_14_TC, excluded)
7. *La cop chase the el hombre* (datal2english, EN_WR_12_30_0.7_14_NC, excluded)
8. *Entonces el hombre montanió el bebé* (datal2english, EN_WR_30_20_3_14_BRB, excluded)

Furthermore, meta-commentaries and first-person statements containing personal opinions dataset were removed prior to annotation to prevent their inclusion in the randomly selected sample. This includes comments referring to the task itself like example (9), as well as first-person singular or plural reflections (e.g. example (10), (11)). These clauses were excluded because they were too infrequent to allow reliable analyses of first-person subject expression or meaningful statistical interpretation. In contrast, Kocher (2025) generally removed such clauses only after assigning ID numbers, resulting in gaps in the annotated dataset. However, their exclusion was not applied consequently as Kocher a small number of first-person clauses still remains in the Kochers annotated dataset (e.g. example (12)), which consequently entered the statistical analyses.

9. [...] *pero [la película] es muy bueno* (datal2english, EN_WR_14_23_1_14_BB, excluded)
10. *I do not know what else to write.* (datal2english, EN_WR_20_21_4_14_PKO, excluded)
11. *Al inicio del short lo vemos vestido mal con la ropa desarreglada* (datal2english, EN_WR_39_62_2_14_MA, excluded)
12. *Nunca aprendí español* (kocher_germanL1_data, DE_WR_20_50_2_14_HO, ID35_2)

After applying Kocher’s (2025) exclusion criteria the final L1 German dataset used for the robustness check and the first close replications contained 2954 clauses while the L1 English dataset consisted of 1285.

The following step consisted of manually annotating the data. Therefore I used Kocher’s annotation scheme and added the two gender related variables **SPEAKERGENDER** and **SUBJECTGENUS** (see Table 2).

Table 2: Variables with annotated values that entered into Kocher’s final model as well as the extended variables **SPEAKERGENDER** (extracted from Kocher (2025): 15) as well as **SUBJECTGENUS** .

Variables	Annotated values
SubjectPosition	PRE, POST, NA (for nullsubj)
ClauseType	MC, SC
InformationStructure	GI(VEN), NE(W), NA (for unknown subject forms)
ReferentialContinuity	Y(ES), N(O), NA (for unknown subject forms)
SubjectType	nullsub, dp, pron
(Proficiency)Score	0-100
Verb	verb of each clause
VerbConstruction	simple-finite, periphrasis-lexical, periphrasis-gerund, modal, infinitive

Variables	Annotated values
SpeakerGender	binary distinction: male, female
SubjectGenus	masc, fem, unknown

Additionally, I added the columns ‘Textnr’, ‘Sentence’, ‘ID’. `SPEAKERGENDER` has already been annotated in Kocher’s (2025) dataset but did not enter into their final model. For the variable `SUBJECTGENUS` I annotated the values *masc* and *fem*, which refer to the binary gender system of Spanish nouns, and *unknown*. The category *unknown* includes cases in which the subject is not clearly identifiable from the prior context or where an impersonal form has been used (see examples (1) - (3)).

1. *Cuando de repente le tiran basura* (datal2german, DE_WR_41_44_30_14_AD, ID98_3)
2. *Por siguiente, se cae algo de arriba* (datal2german, DE_WR_27_21_7_14_SG, ID11_3)
3. *que alguien se olvide [...] el bebe* (datal2german, EN_WR_28_20_7.5_14_EC, ID61_3)

Apart from these variables Kocher (2025) states to have systematically tested the “impact of demographic (and other) factors” in the process of settling on the final models, but that “[t]he models including these variables were not superior on goodness-of-fit measures nor in explanatory power, and therefore these factors are not included in the final models” (2025: 15). Nor in the research paper, nor in Kocher (2025) is specified which factors were originally planned to be integrated into the study. It might be possible that the knowledge of an additional NSL (column `NLSADDITIONAL`) was one of the variables Kocher (2025) originally planned to integrate into the study and then deselected as “the distribution of subject types does not differ significantly between the speakers who know an additional NSL, and those who do not” (2025: 14).

Further, the variable `SPECIFICITY`, that was annotated in Kocher’s (2025) annotation scheme, is not integrated into the final model. For this reason I decided not to annotate it in the L1 English data. Finally, the variable `SENTENCETYPE` that didn’t enter into to the final model hasn’t been explained in the paper nor the codes and also did not enter into Kocher’s final model for which I decided to also not annotate the variable in the L1 German data extension as well as in the L1 English speakers data. The challenges of applying another author’s annotation scheme and of re-running analysis scripts will be discussed in the results section (see Section 5.1.2).

4.3 Replication design

4.3.1 Repetitive research

Repetitive research (Schöch, 2023: 373) refers to the research field that re-examines previously tested hypothesis by using the same data and/or analysis methods as the original studies. As it is a relatively recent area of research that is being explored in a range of different fields, such as linguistics, psychology, marketing, computer science and second language acquisition research (see Röseler et al., 2025: 8). Thus, there exist nearly as many definitions of the concept of reproducibility and replication Barba (n.d.) as fields that are investigating it. Therefore it is necessary to clarify the terminology used in this paper.

First, I seek to study the computational and statistical reproducibility following Stodden’s (2014) approach. **Computational reproducibility** determines “making computational environments available for replication purposes” and refers to providing “information about codes, softwares, hardwares and implementation used” (see The Turing Way Community, 2025). For this work’s purpose I am using a Microsoft Surface Pro 9 running under Windows 11 x64 (build 26200) with the R version 4.6.0 (2026-04-24 ucrt called *Because it was There*) and R Studio version 2026.07.1. Kocher’s (2025) original code that I use for the reproduction part and robustness check (*5_Reproduction_Kocher* and *2_Robustness_Check*), as well as the adapted scripts containing the model extension and testing on the L1 English and L1 German datasets (*3_Extension1_GenderGenus* and *4_Extension2_EnglishGenus*) are provided in `?@sec-code`. Besides, **statistical reproducibility** “repeat[s] results for validation purposes” (Stodden, 2014) and therefore provides detailed information “about the choice of statistical tests, model parameters, and threshold values” (The Turing Way Community, 2025). I provide further information on my study on Kocher (2025) separately in section Section 4.3.2. On the other hand, there is also **inferential reproducibility** which draws “qualitatively similar conclusions from either an independent replication of a study or a reanalysis of the original study” (Goodman et al., 2016: 4). This is especially important as different investigators may draw different conclusions from the same results. My statistical reproduction of Kocher (2025) is accomplished in the discussion of this work `?@sec-discussion`.

Besides these general approaches of reproducible research, there is also the differentiation between results that are *reproducible*, *replicable*, *robust* and *generalizable*. Following Le Foll (2026): 316 I adopt the terminology of The Turing Way Community (2025) to explain the definitions as they come to play in my study (see Figure 1).

The first part of my study is a **reproduction** study testing Kocher’s (2025) original L1 German dataset using the same analysis steps and scripts. Meanwhile, a **replication** uses the same analysis steps on a different dataset and aims “to confirm, consolidate, and extend knowledge and understanding within empirical fields of study by confronting existing claims and ideas using new evidence” (McManus, 2024: 1302). McManus (2024) considers replication to be a kind of continuum, ranging from *exact replication* (without any major changes or manipulation of variables), through *close replication* (1 major change) and *approximate replication* (2 major

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Figure 1: How The Turing Way defines reproducible research (The TuringWayCommunity CC BY 4.0)

changes), to *conceptual replication* (3 major changes). Further, they state that the more manipulation of the study’s set up is made the less appropriate it is for confirming prior study’s results (2024: 1302). Additionally, Figure 1 explains the concept of *robustness* as studies in which “the same dataset is subjected to different analysis workflows to answer the same research question [...] and a qualitatively similar or identical answer is produced” (The Turing Way Community, 2025). Generalisable results are not obtained by applying different analysis steps are implemented on different data, but by “combining replicable and robust findings” (The Turing Way Community, 2025). In the following section I will explain the structure of my reproduction and replication studies.

Prior to evaluate the reproducibility and replicability of Kocher’s (2025) I classify the findability, accessibility, interoperability and reusability of their data and scripts, using the Wilkinson et al.’s (Wilkinson et al., 2016) guiding FAIR principles as orientation (see Figure 2). The evaluation is provided in Section 5.1.1.

In order to make my data and scripts *findable* as possible I share them in [see ?@sec-appendix-C thesis_Tizianallie

accessible

interoperable

reusable

4.3.2 Statistic modelling

reproduction of Kocher (2025)

I conduct *two close replication* studies by, first, adding gender as a new variable to the model and the extended L1 German dataset. In the second close replication, I apply the new model to the L1 English speaker dataset. This allows for comparisons not only of the role that gender plays in L1 German learner data, but also of the impact of the original and additional

Box 2 The FAIR Guiding Principles
To be Findable: F1. (meta)data are assigned a globally unique and persistent identifier F2. data are described with rich metadata (defined by R1 below) F3. metadata clearly and explicitly include the identifier of the data it describes F4. (meta)data are registered or indexed in a searchable resource To be Accessible: A1. (meta)data are retrievable by their identifier using a standardized communications protocol A1.1 the protocol is open, free, and universally implementable A1.2 the protocol allows for an authentication and authorization procedure, where necessary A2. metadata are accessible, even when the data are no longer available To be Interoperable: I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation. I2. (meta)data use vocabularies that follow FAIR principles I3. (meta)data include qualified references to other (meta)data To be Reusable: R1. (meta)data are richly described with a plurality of accurate and relevant attributes R1.1. (meta)data are released with a clear and accessible data usage license R1.2. (meta)data are associated with detailed provenance R1.3. (meta)data meet domain-relevant community standards

Figure 2: List of Wilkinson et al.’s FAIR Principles

variables on a group of learners with a comparable non-NSL. Therefore, “reproduction and replication should always be considered together and if possible, *reproduction should come before replication*” (Röseler et al., 2025: 9).

I decided to broaden the use of the term *robustness* to the robustness check that I am conducting by adding new participants to the L1 German dataset.⁴ Finally, I also test the robustness of the manual annotation by re-annotating the sample of three speaker texts.

Reproduction of Kocher (2025)

- statistical tests
- model parameters
- threshold values
- Demographics
- Descriptive
- MuPDARF design
- Variable importance (not mentioned in detail)

– Modell 1: L1Spanisch – Referenzmodell (Random Forest / MuPDARF)

⁴Using the term *robustness check* for repeating the analysis with a slightly extended dataset and for manually re-annotating a subset of Kocher’s dataset (2025) is fairly debatable. Although extending the dataset with a small number of additional data could also be viewed as a form of replication, as the analysis remains the same, its primary purpose is to assess the robustness of the results rather than to replicate them using completely different data.

- Modell 2a: Anwendung des Modells 1 auf L2 Deutsch
- Modell 3a: Vergleich der L1 Modellvorhersagen mit den tatsächlichen L2 Daten
- Modell 4a: Target-likeness Modell

L1 German extension: The role of the speaker's gender and subject genus

XXX

L1 English extension: The role of the speaker's gender and subject genus

- Modell 2b: Anwendung des Modells 1 auf L2 Englisch
- Modell 3b: Vergleich der L1 Modellvorhersagen mit den tatsächlichen L2 Daten
- Modell 4b: Target-likeness model

4.4 Model accuracy and model assumptions

and how I will measure that!

- how I will measure that!
 - package used (see elen textbook chpt 13) > {performance} package
- thresholds in collinearity, multicollinearity?

5 Results

5.1 Reproduction of Kocher's original study

5.1.1 How FAIR is Kocher's study?

XXXTableXXX

5.1.2 Computational and statistical reproduction

- Computational repro of the results:
 - Bestätigung von Kochers descriptive and demographic values
 - mupdarf successful?
 - Model accuracy and model assumptions (see Elens textbook chpt 13)

5.1.3 Model accuracy and model assumptions

- (z.B. Multicollinearity) and how I will measure that!

```
::: {.cell}
```

```
:::
```

```
::: {.cell}
```

```
:::
```

- Reproduction of tree successful??

Findable, Accessible, Interoperable, Reusable (FAIR) erfüllt?

“Then we discuss the challenges of applying another author’s annotation scheme and of re-running analysis scripts, thus demonstrating that achieving reproducibility in LCR studies is not as trivial as it might appear.”

- No scripts containing information about preprocessing the NSLHOME and NLSADDITIONAL entries available.
- Also not explained how each corresponding clause was annotated (what counts as a clause, why did Kocher opt for the categories they classified e.g. the VERBCONSTRUCTION column by)
- The packages themselves but no package versions are mentioned in the paper
- Some cases, that Kocher actually claims to have excluded pre-annotation, are still within their annotated data (see Section 4.2.2).
- Ist die Reproduktion, die Darstellung von Figure 4, Table 8, Figure 5, und insbesondere Figure 6 gelungen

5.2 Robustness check

5.2.1 Robustness of Kocher’s annotation

- Robustness check by annotating a sample of Kocher’s L1 German speaker data

5.2.2 Descriptive statistics of the extended L1 German speaker data

- Robustness of Kocher's L1 German speaker results by extending the dataset
- Computational (**numbers and values**) vs. inferential reproduction (**interpreting results**) of Kocher (2025)

5.3 Close replication 1: Gender model on L1 German data

- only Mupdarf

5.4 Close replication 2: Gender model on L1 English data

5.4.1 Descriptive statistics of the L1 English speaker data

- In the L1 English speaker data the distribution of subject types differs significantly between the speakers who know an additional NSL, and those who do not. In Kocher's L1 German speaker data this is not the case! zunächst Gesamtverteilung, dann modellbasierter Vergleich (Ergebnisse aus Modell 6 mit Haupteffekte, Interaktionen, Effektgröße)
- descriptive stats & demographics

5.4.2 MuPDARF of the L1 English speaker data

- mupdarf that compares results to Spanish native data only!

6 Discussion

XXXEinleitungstextXXX

6.1 Interpreting the role of the speaker's gender and subject genus as part of the subject realisation in L2 Spanish

6.1.1 L1 German

- just interpreting results in comparison to Spanish natives!

6.1.2 L1 English

- just interpreting results in comparison to Spanish natives!

6.2 Comparison L1 German vs. L1 Englisch

- Unterschiedliches Vorgehen bei Transfer? Feature-Reassembly-Komplexität unterschiedlich?

6.3 Theoretical implications

- Bedeutung der Ergebnisse für die Interface Hypothesis vs. Feature Reassembly Hypothesis

7 Limitations

- evtl. Komplikationen bei der Operationalisierung der Variablen
- Ungleichverteilung der Language Proficiency im Datensatz (right skewed!)
- Auch wenn sich das Korpus gut eignet für sprachübergreifende Vergleichstudien, lassen sich die ergebnisse nicht weiter generalisieren sondern nur auf written learner language data beziehen

8 Conclusion and Outlook

Beantwortung der Forschungsfragen Beitrag zur Forschung zur Subjektrealisierung L2-Spanisch Sprecher:innen des L1 Deutschen und Englischen * Perspektiven für weitere Forschung

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9.2 Packages Used

R version 4.6.0 (2026-04-24 ucrt)

Platform: x86_64-w64-mingw32/x64

Running under: Windows 11 x64 (build 26200)

Matrix products: default

LAPACK version 3.12.1

locale:

[1] LC_COLLATE=German_Germany.utf8 LC_CTYPE=German_Germany.utf8

[3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C

[5] LC_TIME=German_Germany.utf8

time zone: Europe/Berlin

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] lubridate_1.9.5 forcats_1.0.1 stringr_1.6.0 dplyr_1.2.1

[5] purrr_1.2.2 readr_2.2.0 tidyr_1.3.2 tibble_3.3.1

[9] tidyverse_2.0.0 here_1.0.2 knitr_1.51 ggplot2_4.0.3

loaded via a namespace (and not attached):

[1] gtable_0.3.6 jsonlite_2.0.0 compiler_4.6.0 tidyselect_1.2.1

[5] scales_1.4.0 fastmap_1.2.0 R6_2.6.1 generics_0.1.4

[9] rprojroot_2.1.1 pillar_1.11.1 RColorBrewer_1.1-3 tzdb_0.5.0

[13] rlang_1.2.0 stringi_1.8.7 xfun_0.57 S7_0.2.2

[17] otel_0.2.0 timechange_0.4.0 cli_3.6.6 withr_3.0.2

[21] magrittr_2.0.5 digest_0.6.39 grid_4.6.0 rstudioapi_0.18.0

[25] hms_1.1.4 lifecycle_1.0.5 vctrs_0.7.3 evaluate_1.0.5

[29] glue_1.8.1 farver_2.1.2 rmarkdown_2.31 tools_4.6.0

[33] pkgconfig_2.0.3 htmltools_0.5.9

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Appendix A: Additional Tables

Add additional tables, if you like.

Appendix B: Additional Figures

Add additional figures, if you like.

Appendix C: Code

The complete codes used for the analyses in this thesis are contained within the Quarto (.qmd) files, which is accessible via the following Git repository: **LINK**